

GAUTAM SINGH JYALA

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PUBLICATION

Gautam Singh Jyala, Rahul Chauhan, Ruchira Rawat, (2024), "*An Imitation of Humanoid Robot: A Qualitative Analysis*" in Challenges in Information, Communication, and Computing Technology, 1st Edition, CRC Press. [Link](#)

EDUCATION

Graphic Era Hill University B.Tech in Computer Science - 8.15 CGPA

2022 - 2026

PROJECTS

GenAI Crypto Research Assistant (RAG-Based System)

- Built a **domain-specific RAG assistant** integrating curated data and real-time APIs, reducing hallucinations by **30%+**.
- Implemented a **low-latency pipeline** with **ChromaDB** and **MiniLM embeddings**, achieving **sub-500ms retrieval**.
- Developed a modular **FastAPI backend** with optional **Streamlit UI**, enabling seamless dataset/API swaps.
- Validated the system on **1k+ crypto documents** using **LangChain** and **OpenRouter LLMs**.

Email/SMS Spam Classifier Web App

[Live](#)

- Built a spam detector using a **Stacked Ensemble Model** (SVM, NB, ExtraTrees, RF) with **96.8% accuracy** and **96.5% precision**.
- Processed **5,500+ messages** with tokenization, stemming, stopword removal, and TF-IDF (3000 features).
- Achieved a **2x precision gain** using **5-fold stacking** over baseline models.
- Deployed on **Streamlit Cloud** with a **lightweight <2MB model** for real-time predictions.

Banana Life Span Prediction

[GitHub](#)

- Built a **TensorFlow + MobileNetV2** model to classify **5 ripeness stages** and predict **0–6 eatable days** of bananas.
- Trained with **20% validation split**, **data augmentation**, and **20 fine-tuning epochs**, achieving **92%+ validation** and **90%+ test accuracy**.
- Designed a ripeness-to-days mapping system for **real-world food quality monitoring**.
- Visualized metrics (accuracy/loss) and exported the model as `banana_quality_model.keras` for deployment.

TECHNICAL SKILLS

- **Languages:** C, C++, Java, Python
- **Libraries:** Numpy, Pandas, Matplotlib, SKlearn, Seaborn
- **Database:** MySQL
- **Coursework:** OOPS, DBMS, Data Structures & Algorithms

ACHIEVEMENTS

- Played North-Zone(AIU)Nationals in Badminton (2025).
- Solved **350+ DSA problems** on Leetcode and GFG.
- Awarded Leetcode Annual Badge 2023 for being in **top 0.4% globally**.
- Earned 100 and 200 Days Leetcode Streak Badges in 2024 and 2025.